

Seasonal variations in *Schistosoma haematobium* egg excretion in school-age girls in rural KwaZulu-Natal Province, South Africa.

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A predominant feature of *Schistosoma haematobium* infection is urinary egg excretion, and microscopic egg detection remains the accepted standard field diagnostic tool. Praziquantel is the drug of choice for schistosomiasis, and the World Health Organization recommends that it should be administered to all children >4 years of age living in schistosomiasis-endemic areas. The frequency of mass drug administration depends on the prevalence rate in the community. Urinary schistosome egg output has a day-to-day and hour-to-hour intrasubject variation. Therefore, it is important to assess possible seasonal variations in egg excretion to improve the planning of drug treatment. To assess the influence of seasonality on urinary schistosome egg excretion in South Africa (SA). We performed a prospective cohort study, exploring seasonal variations of *S. haematobium* egg excretion in 184 girls aged 10 - 12 years from randomly selected schools in a rural area of KwaZulu-Natal Province, SA. The area has a subtropical climate characterised by a cool dry season and a hot humid season. For children, water contact is higher in the latter season. At baseline, 108 girls were examined in the hot season, and 76 in the cold season. In the next year's cold season the untreated patients were re-investigated before treatment. There was a decrease in infection in the group initially tested in the hot season compared with the group tested in the cold season at both time points when adjusted for age and water contact (adjusted odds ratio 3.61 (95% confidence interval 1.14 - 11.44); $p=0.03$). This unique study shows that schistosomiasis prevalence determined by microscopy exhibits seasonal variation, with a higher prevalence in the hot rainy season. Precise community prevalence estimations are key in decisions to treat communities. There was significantly lower egg output in the cold season, and sampling in that season may therefore underestimate the prevalence of urinary schistosomiasis. The study indicates that sampling in SA should be done in the hot season.

Microglia target synaptic sites early during excitatory circuit disassembly in neurodegeneration.

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During development, microglia prune excess synapses to refine neuronal circuits. In neurodegeneration, the role of microglia-mediated synaptic pruning in circuit remodeling and dysfunction is important for developing therapies aimed at modulating microglial function. Here we analyzed the role of microglia in the synapse disassembly of degenerating postsynaptic neurons in the inner retina. After inducing transient intraocular pressure elevation to injure retinal ganglion cells, microglia increase in number, shift to amoeboid morphology, and exhibit greater process movement. Furthermore, due to the greater number of microglia, there is increased colocalization of microglia with synaptic components throughout the inner plexiform layer and with excitatory synaptic sites along individual ganglion cell dendrites. Microglia depletion partially restores ganglion cell function, suggesting that microglia activation may be neurotoxic in early neurodegeneration. Our results demonstrate the important role of microglia in synapse disassembly in degenerating circuits, highlighting their recruitment to synaptic sites early after neuronal injury.

Role of microglia in brain development after viral infection.

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Microglia are immune cells in the brain that originate from the yolk sac and enter the developing brain before birth. They play critical roles in brain development by supporting neural precursor proliferation, synaptic pruning, and circuit formation. However, microglia are also vulnerable to environmental factors, such as infection and stress that may alter their phenotype and function. Viral infection activates microglia to produce inflammatory cytokines and anti-viral responses that protect the brain from damage. However, excessive or prolonged microglial activation impairs brain development and leads to long-term consequences such as autism spectrum disorder and schizophrenia spectrum disorder. Moreover, certain viruses may attack microglia and deploy them as "Trojan horses" to infiltrate the brain. In this brief review, we describe the function of microglia during brain development and examine their roles after infection through microglia-neural crosstalk. We also identify limitations for current studies and highlight future investigated questions.

Synaptic pruning by microglia is necessary for normal brain development.

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Microglia are highly motile phagocytic cells that infiltrate and take up residence in the developing brain, where they are thought to provide a surveillance and scavenging function. However, although microglia have been shown to engulf and clear damaged cellular debris after brain insult, it remains less clear what role microglia play in the uninjured brain. Here, we show that microglia actively engulf synaptic material and play a major role in synaptic pruning during postnatal development in mice. These findings link microglia surveillance to synaptic maturation and suggest that deficits in microglia function may contribute to synaptic abnormalities seen in some neurodevelopmental disorders.

Molecular mechanisms underlying microglial sensing and phagocytosis in synaptic pruning.

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Microglia are the main non-neuronal cells in the central nervous system that have important roles in brain development and functional connectivity of neural circuits. In brain physiology, highly dynamic microglial processes are facilitated to sense the surrounding environment and stimuli. Once the brain switches its functional states, microglia are recruited to specific sites to exert their immune functions, including the release of cytokines and phagocytosis of cellular debris. The crosstalk of microglia between neurons, neural stem cells, endothelial cells, oligodendrocytes, and astrocytes contributes to their functions in synapse pruning, neurogenesis, vascularization, myelination, and blood-brain barrier permeability. In this review, we highlight the neuron-derived "find-me," "eat-me," and "don't eat-me" molecular signals that drive microglia in response to changes in neuronal activity for synapse refinement during brain development. This review reveals the molecular mechanism of neuron-microglia interaction in synaptic pruning and presents novel ideas for the synaptic pruning of microglia in disease, thereby providing important clues for discovery of target drugs and development of nervous system disease treatment methods targeting synaptic dysfunction.

Microglia sculpt postnatal neural circuits in an activity and complement-dependent manner.

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Microglia are the resident CNS immune cells and active surveyors of the extracellular environment. While past work has focused on the role of these cells during disease, recent imaging studies reveal dynamic interactions between microglia and synaptic elements in the healthy brain. Despite these intriguing observations, the precise function of microglia at remodeling synapses and the mechanisms that underlie microglia-synapse interactions remain elusive. In the current study, we demonstrate a role for microglia in activity-dependent synaptic pruning in the postnatal retinogeniculate system. We show that microglia engulf presynaptic inputs during peak retinogeniculate pruning and that engulfment is dependent upon neural activity and the microglia-specific phagocytic signaling pathway, complement receptor 3(CR3)/C3. Furthermore, disrupting microglia-specific CR3/C3 signaling resulted in sustained deficits in synaptic connectivity. These results define a role for microglia during postnatal development and identify underlying mechanisms by which microglia engulf and remodel developing synapses.

Synaptic Pruning by Microglia: Lessons from Genetic Studies in Mice.

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Neural circuits are subjected to refinement throughout life. The dynamic addition and elimination (pruning) of synapses are necessary for maturation of neural circuits and synaptic plasticity. Due to their phagocytic nature, microglia have been considered as the primary mediators of synaptic pruning. Synaptic pruning can strengthen an active synapse by removing excess weaker synapses during development. Inappropriate synaptic pruning can often influence a disease outcome or an injury response. This review offers a focused discussion on microglial roles in synaptic pruning, based on the evidence gathered from genetic manipulations in mice. Genetically labeled microglia and synapses often allow assessment of their interactions in real time. Further manipulations involving synaptically localized molecules, neuronally or glial-derived diffusible factors, and their respective cognate receptors in microglia provide critical evidence in support of a direct role of microglia in synaptic pruning. We discuss microglial contact-dependent "eat-me," "don't-eat-me," and "find-me" signals, as well as recently identified noncontact pruning, under the contexts of neural circuit, brain region, developmental window, and an injury or a disease state.

Microglia-mediated synaptic pruning as a key deficit in neurodevelopmental disorders: Hype or hope?

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There is a consensus in the field that microglia play a prominent role in neurodevelopmental processes like synaptic pruning and neuronal network maturation. Thus, a current momentum of associating microglia deficits with neurodevelopmental disorders (NDDs) emerged. This concept is challenged by rodent studies and clinical data. Intriguingly, reduced numbers of microglia or altered microglial functions

do not necessarily lead to overt NDD phenotypes, and neuropsychiatric symptoms seem to develop primarily in adulthood. Hence, it remains open for discussion whether microglia are truly indispensable for healthy neurodevelopment. Here, we critically discuss the role of microglia in synaptic pruning and highlight area- and age dependency. We propose an updated model of microglia-mediated synaptic pruning in the context of NDDs and discuss the potential of targeting microglia for treatment of these disorders.

Deficient neuron-microglia signaling results in impaired functional brain connectivity and social behavior.

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Microglia are phagocytic cells that infiltrate the brain during development and have a role in the elimination of synapses during brain maturation. Changes in microglial morphology and gene expression have been associated with neurodevelopmental disorders. However, it remains unknown whether these changes are a primary cause or a secondary consequence of neuronal deficits. Here we tested whether a primary deficit in microglia was sufficient to induce some autism-related behavioral and functional connectivity deficits. Mice lacking the chemokine receptor Cx3cr1 exhibit a transient reduction of microglia during the early postnatal period and a consequent deficit in synaptic pruning. We show that deficient synaptic pruning is associated with weak synaptic transmission, decreased functional brain connectivity, deficits in social interaction and increased repetitive-behavior phenotypes that have been previously associated with autism and other neurodevelopmental and neuropsychiatric disorders. These findings open the possibility that disruptions in microglia-mediated synaptic pruning could contribute to neurodevelopmental and neuropsychiatric disorders.

Microglia during development and aging.

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Microglia are critical nervous system-specific cells influencing brain development, maintenance of the neural environment, response to injury, and repair. They contribute to neuronal proliferation and differentiation, pruning of dying neurons, synaptic remodeling and clearance of debris and aberrant proteins. Colonization of the brain occurs during gestation with an expansion following birth with localization stimulated by programmed neuronal death, synaptic pruning, and axonal degeneration. Changes in microglia phenotype relate to cellular processes including specific neurotransmitter, pattern recognition, or immune-related receptor activation. Upon activation, microglia cells have the capacity to release a number of substances, e.g., cytokines, chemokines, nitric oxide, and reactive oxygen species, which could be detrimental or beneficial to the surrounding cells. With aging, microglia shift their morphology and may display diminished capacity for normal functions related to migration, clearance, and the ability to shift from a pro-inflammatory to an anti-inflammatory state to regulate injury and repair. This shift in microglia potentially contributes to increased susceptibility and neurodegeneration as a function of age. In the current review, information is provided on the colonization of the brain by microglia, the expression of various pattern recognition receptors to regulate migration and phagocytosis, and the shift in related functions that occur in normal aging.